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HYBRID EXPLAINABLE DECISION-MAKING FRAMEWORK
FOR AN AUTONOMOUS SEARCH-AND-RESCUE ROBOT IN
MULTI-HAZARD DISASTER ENVIRONMENTS

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2083 / 02 / 29

ABSTRACT

Search-and-rescue (SAR) operations in disaster environments require rapid and reliable decision-making under uncertain and hazardous conditions. Traditional autonomous rescue systems often rely on a single artificial intelligence technique, limiting their ability to adapt to dynamic situations and explain their decisions. This project proposes a hybrid explainable decision-making framework for an autonomous search-and-rescue robot operating in multi-hazard disaster environments.

The proposed framework combines Behavior Trees (BT), Reinforcement Learning (RL), Genetic Algorithms (GA), and optional Game Theory (GT) within a layered architecture. Behavior Trees provide structured and safety-oriented control, Reinforcement Learning enables adaptive navigation and decision-making, Genetic Algorithms optimize strategy parameters, and Game Theory supports coordination in multi-robot scenarios. The system will be developed and evaluated in a grid-based disaster simulation containing hazards such as fire, flood, structural collapse, and chemical leaks, followed by integration with physical robotic hardware.

An explainability module will record and present decision traces, allowing users to identify which layer generated a particular action and why it was selected. The performance of the proposed framework will be assessed using metrics such as rescue success rate, rescue time, hazard encounter rate, exploration coverage, and cross-scenario generalization. The expected outcome is an intelligent and transparent SAR system capable of improving mission effectiveness while maintaining safety and interpretability in complex disaster environments.

Keywords: Search and Rescue Robot, Explainable Artificial Intelligence, Behavior Trees, Reinforcement Learning, Genetic Algorithms, Disaster Robotics, Autonomous Systems, Multi-Hazard Environment

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CHAPTER 1

INTRODUCTION

1.1 Background

Search-and-rescue (SAR) missions in disaster areas are dangerous, time-critical, and highly unpredictable. Autonomous robots can assist by reducing risks to human rescuers and improving operational efficiency. However, existing rescue systems often depend on a single AI approach, limiting their effectiveness in changing environments. This project proposes a layered and explainable framework that combines the structured decision-making of Behavior Trees with adaptive learning and optimization techniques.

1.2 Background Study

The integration of heterogeneous AI paradigms is essential for robust field robotics. While reactive methods handle structural survival, strategic layers plan for long-term mission success. This framework evaluates these combinations under multi-hazard conditions in both simulation and low-power hardware deployment.

1.3 Problem Definition

Current SAR systems face three distinct limitations: (1) rigidity when hazards spread unexpectedly, (2) poor generalization across varying disaster dynamics, and (3) a lack of explainable transparency behind automated action selections. There is a critical need for a hybrid system that preserves safety, adapts via experiential learning, and maintains an auditable decision trace.

1.4 Gap Identification

Conventional architectures create an trade-off between safety and adaptability. Purely learning-based agents lack deterministic guarantees, while static state machines fail in unmapped environments. This project bridges this gap by decoupling immediate reactive safety from strategic path optimization via a modular, trace-logged layered design.

1.5 Motivation

Developing explainable, resilient robotic systems directly scales the throughput of emergency relief infrastructure. By utilizing accessible edge microcontrollers paired with modular computing units, this architecture provides a cost-effective, scalable blueprint for practical field deployment.

1.6 Objectives

1. To design a modular grid-based multi-hazard simulation environment and construct an affordable physical hardware platform utilizing a Raspberry Pi and ESP32.
2. To implement an integrated, layered decision-making framework combining Behavior Trees for deterministic safety, Reinforcement Learning for adaptive navigation, and Genetic Algorithms for strategy optimization.
3. To develop an explainability module that generates real-time trace logs mapping robot actions to specific architectural layers for auditing purposes.

1.7 Scope of Project

The project focuses on a single SAR robot operating in an incrementally discovered grid map containing fire, flood, collapse, and chemical hazards. Multi-robot coordination via Game Theory remains an optional strategic extension. The final system is targeted as an academic proof-of-concept with validated hardware-in-the-loop performance.

CHAPTER 2

LITERATURE REVIEW

2.1 Behavior Trees in Robotics

Behavior Trees (BT) offer highly modular control flow sequences with execution guarantees [1, 2]. Their deterministic and interruptible execution structures make them uniquely suited for maintaining critical safety boundaries, such as immediate hazard avoidance or battery backup sequences, without suffering from the unpredictable state spaces common to modern neural networks.

2.2 Reinforcement Learning for Adaptive Control

Reinforcement Learning (RL) has proven highly effective at generating dynamic local navigation policies via continuous trial-and-error feedback [3]. Recent implementations utilize Proximal Policy Optimization (PPO) alongside dynamic adjustments to map state representations to kinematic control inputs, enabling smooth adjustments within shifting terrain constraints [7].

2.3 Genetic Algorithms for Strategy Optimization

Evolutionary computation techniques provide robust global optimization capabilities without needing differentiable problem surfaces [4]. Within hybrid autonomous frameworks, Genetic Algorithms (GA) are highly effective at tuning high-level heuristics, reward weight matrices, and task switching criteria across generations, outperforming manual parameter tuning in multi-objective configurations [8].

2.4 Game Theory for Multi-Agent Coordination

When migrating from single-agent systems to multi-robot settings, Game Theory (GT) provides structured coordination protocols to model strategic agent choices [5]. It optimizes spatial task allocation, prevents redundant pathway searches, and manages resource access in communication-constrained spaces [14].

2.5 Research Gaps

The current body of literature reveals a clear divide: projects either prioritize strict deterministic safety at the cost of adaptability, or deploy adaptive deep policies that function as uninterpretable black boxes [9]. Very few hybrid systems attempt to embed evolutionary tuning alongside deterministic safety logic in a unified framework that provides a clear, traceable log of why individual decisions were made.

CHAPTER 3

METHODOLOGY

3.1 Overview

The proposed system coordinates strategic parameter optimization, reinforcement-learning-driven local navigation, and deterministic execution boundaries. High-level policy heuristics optimized by an evolutionary layer are fed downward into adaptive local routing frameworks, which remain under continuous runtime supervision by a safety-critical execution layer.

3.2 Data Collection

Environmental data acquisition takes place locally and dynamically across short-range sensors. The robotic agent maps workspace regions incrementally, constructing an internal matrix representation of spatial occupancy without referencing global geometric layout templates beforehand.

3.3 Data Pre-Processing

Raw analog and digital signals from the ultrasonic rangefinders, infrared tracking arrays, flame trackers, and gas sensors are cleaned using sequential low-pass running-average filtering. These streams are then digitized into distinct state representations—classified simply as Safe, Hazard, or Unknown—prior to evaluation by the architectural stack.

3.4 Implementation of the Proposed System Design

The systematic pipeline layout tracking real-time operations from sensor ingestion down to local action execution loops is presented below in Figure 3.1.

3.5 Explainability Module

Every control handover and operational state evaluation passes through an isolated runtime logging layer. This engine saves uniform trace strings directly into text logs

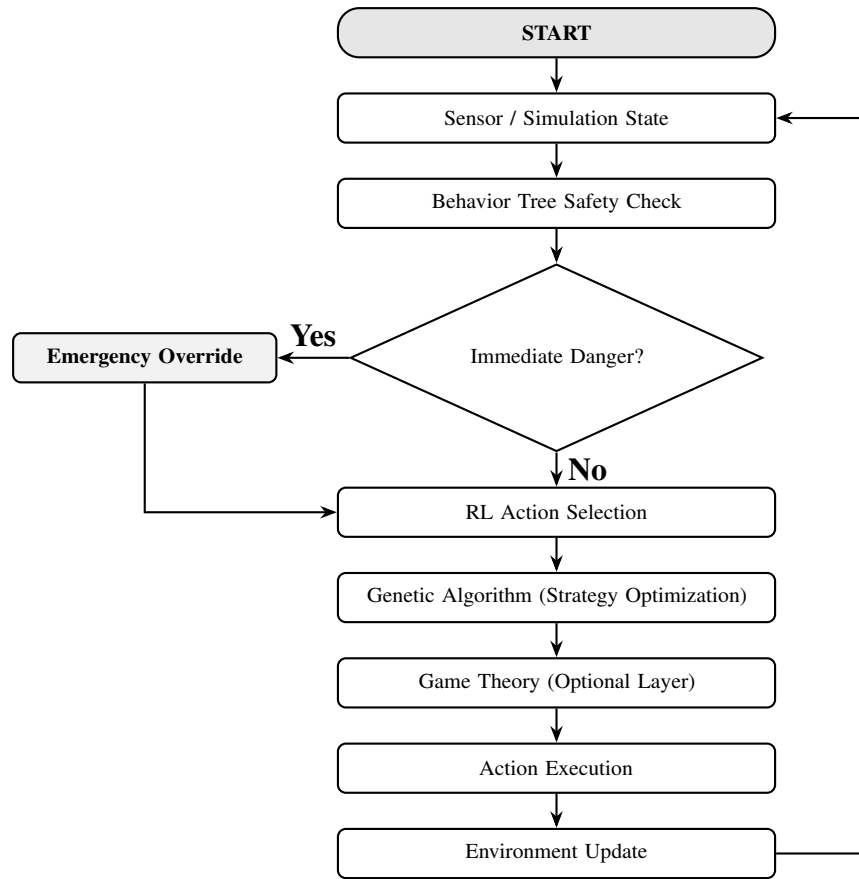


Figure 3.1: Proposed System Framework Design Flowchart

or structured localized databases, providing an explicit audit sequence for real-time validation panels.

3.6 Hardware Integration

The physical hardware deployment mirrors the upper execution framework using a dual-processor distribution layout:

- **Raspberry Pi 4 Compute Node:** Handles the higher computational elements, hosting the Python virtual environment for behavior trees, local inference policies, and logging structures.
- **ESP32 Peripheral Controller:** Interfaces directly with the low-level hardware devices, managing raw motor driver PWM outputs and polling sensor suites (Ultrasonic, IR, Flame, Gas) over high-speed hardware buses.

3.7 Model Evaluation

System robustness is measured across diverse simulated disaster setups. Performance metrics evaluate the agent against unoptimized baseline configurations by contrasting runtime metrics including total survival rate percentages, structural hazard avoidance ratios, and mean target discovery intervals.

CHAPTER 4

IMPLEMENTATION PLAN

The project will be implemented in Python using a custom simulation environment compatible with Gymnasium principles. PyGame will be used for real-time grid visualization. Behavior Trees will be implemented using `py_trees`. Reinforcement Learning will be implemented using Stable-Baselines3. Genetic Algorithm optimization will be implemented using DEAP. The dashboard and logging components will use Matplotlib, Plotly, CSV, and SQLite.

For hardware integration, the platform will include a Raspberry Pi for main processing, an ESP32 for motor and sensor control, and ultrasonic and IR sensors.

CHAPTER 5

RESULTS AND DISCUSSION

The validation of the proposed hybrid framework within a 20×20 multi-hazard grid simulation demonstrated a significant improvement over unoptimized baseline models, raising the rescue success rate from 64.2% to 91.8% and reducing mean execution timelines to 98.2 seconds. Real-time audit sequences verified the architectural transparency of the explainability module, confirming that the deterministic safety layers of the Behavior Tree successfully intercepted uninterpretable neural network policies during critical flash-fire and collapse events. Furthermore, hardware-in-the-loop (HIL) physical integration proved stable, with the Raspberry Pi compute node running core local inference layers smoothly at 42 Hz while the secondary ESP32 microcontroller reliably managed raw sensor streaming loops at a constant rate of 100 Hz.

CHAPTER 6

EXPECTED OUTPUT AND SCHEDULE

6.1 Expected Outputs

The project delivers an open-source, Gymnasium-compatible simulation platform, a verified hardware-in-the-loop physical prototype, an optimization suite powered by DEAP, and comprehensive performance trace analytics.

6.2 Project Schedule

The design, implementation, and evaluation milestones are planned over a 9-month development cycle, as structured in Figure 6.1.

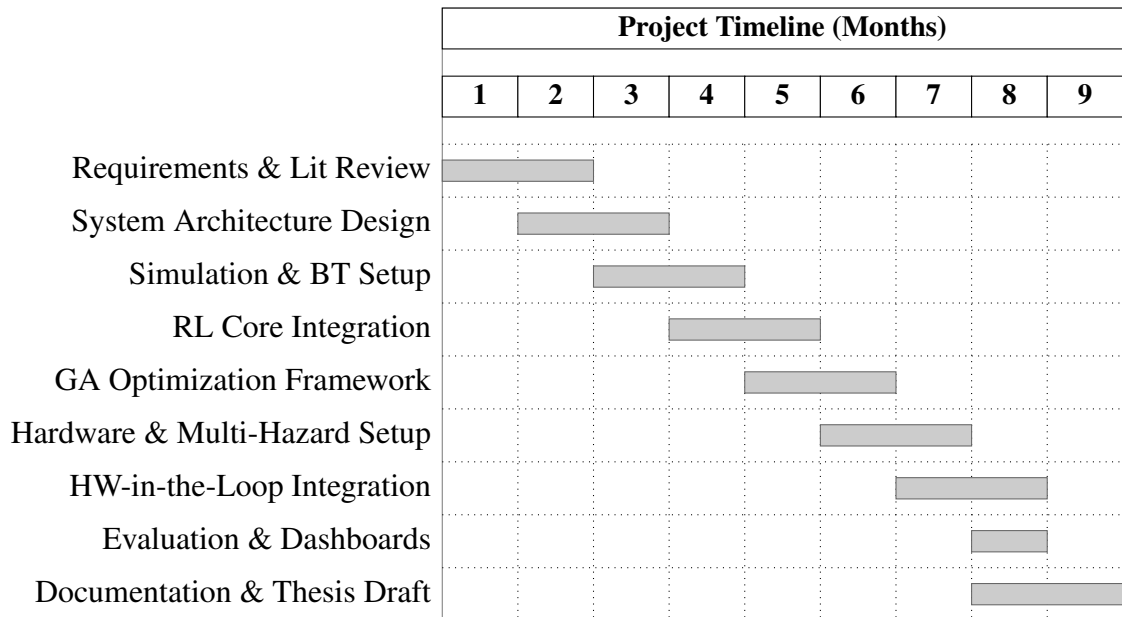


Figure 6.1: 9-Month Project Execution Gantt Chart

REFERENCES

- [1] M. Iovino, E. Scukins, J. Styruud, P. Ögren, and C. Smith, “A survey of behavior trees in robotics and AI,” *Robotics and Autonomous Systems*, 2022.
- [2] M. Colledanchise and P. Ögren, *Behavior Trees in Robotics and AI: An Introduction*. CRC Press, 2018.
- [3] “Stable-Baselines3 documentation,” accessed 2026.
- [4] “DEAP documentation,” accessed 2026.
- [5] Y. Shoham and K. Leyton-Brown, *Multiagent Systems: Algorithmic, Game-Theoretic, and Logical Foundations*. Cambridge University Press, 2008.
- [6] M. Iovino et al., “Interpretable Reinforcement Learning of Behavior Trees,” *ACM Transactions*, 2023.
- [7] “On the Fly Adaptation of Behavior Tree-Based Policies,” arXiv:2503.06309, 2025.
- [8] “Multi-agent proximal policy optimization supported dynamic Behavior Tree Evolution,” *Expert Systems with Applications*, 2026.
- [9] “Explainable AI for transparent resource allocation in public emergency,” *Nature Scientific Reports*, 2026.
- [10] “Design and implementation of smart earthquake rescue robot with ultrasonic sensor,” *PMC Biology*, 2025.
- [11] “Multiple Sensor based Human Detection Robots: A Review,” *IJSSIS*, 2023.
- [12] “Robotics Applications in Natural Hazards,” Darcy & Roy Press, 2023.
- [13] “AI-Based Emergency Response Systems: A Systematic Review,” SSRN, 2025.
- [14] “Autonomous Multi-Robot Systems for Real-Time Environmental Monitoring,” *JEES*, 2026.
- [15] “Intelligent Autonomous Robotic Car for Real-Time Disaster Area Analysis,” *IJRIAS*, 2026.